Grading and Feedback

Course: REI603M The AI Lifecycle
Assignment: Assignment 3 - Evaluation Dataset & Prompt Engineering
Students: jre5, sos106, sbs87
Project: AI Math Coach

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Grade: 10/10

The submission meets or exceeds every requirement – well-structured dataset, four iterated prompt versions with documented rationale, Pydantic-enforced structured outputs, systematic evaluation with error analysis, and a complete interaction data plan – so no deductions apply.

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Checklist & Grade Computation

Part 1: Evaluation Dataset

- [x] At least 50 test cases – exactly 50 image-based test cases in `assignment3.csv` with corresponding images in `img/`
- [x] Clear categories/dimensions – categories include: fully solved check, correct-so-far hint, incorrect hint, incorrect check, reveal, edge cases, ambiguities, and answer leakage scenarios. Error types are also annotated (derivative error, sign error, log rule error, etc.)
- [x] Dataset creation methodology documented – `readme.md` and `version_notes.md` describe the process; slides 3-4 cover design principles
- [x] Ground truth for each test case – `expected_verdict` column present for all 50 rows, plus `expected_behaviour` and `error_type`
- [x] Edge cases included – items 42-50 cover crossing values in equations, formerkjakoennun for inequalities, graphical solutions, hint when fully solved, multiple solution methods, multiple errors, unclear handwriting, and answer leakage attempts
- [x] Work split across group members (author field) – author column present with entries for "Johannes" and "Solvi"; third member likely contributed to code, prompts, and presentation

Part 2: Prompt Engineering

- [x] Multiple prompting strategies – v1 (zero-shot baseline), v2 (expanded instruction with explicit reasoning chain), v3 (few-shot with curated examples), v4 (expanded few-shot with stricter policy and error taxonomy)
- [x] Structured outputs used – Pydantic `BaseModel` (`LLMResponse`) enforcing `verdict`, `response_type`, `message_is` schema; Gemini API `response_json_schema` parameter used
- [x] Iterative refinement documented – `version_notes.md` provides a detailed table of what changed, why, and observed impact for each of the four versions
- [x] All prompts included in submission – `prompts/v1.txt` through `prompts/v4.txt` all present

Part 3: Experimentation & Analysis

- [x] Systematic evaluation on full dataset – all 4 prompt versions evaluated on all 50 test cases (200 total runs), results saved as `results_v1.csv` through `results_v4.csv`
- [x] Quantitative or qualitative metrics – quantitative: verdict accuracy (94-96%) and non-feasible ratio (0-12%); qualitative: correctness rate, hint usefulness (1-5), clarity (1-5), policy violations, answer leakage detection
- [x] Error analysis with categorized failures – slide 8 identifies persistent errors (ambiguity misclassification, false negatives on dense symbolic solutions) and version-specific failures (v2 policy violations); `qualitative_review.py` implements heuristic scoring with detailed rationale

- [x] Reflection – slide 9 covers what worked (structured outputs, few-shot), what surprised (longer prompts reduced compliance), evaluation gaps (limited unclear cases, limited handwriting variability), and concrete next steps

Part 4: Interaction Data Planning

- [x] Usage patterns – core events defined: `problem_started`, `step_submitted`, `mode_requested`, `llm_response_generated`, `user_retry`, `reveal_clicked`, `problem_completed`, `session_abandoned`
- [x] Quality signals – valid next step within two actions after hint, correction after first-error feedback, user ratings (thumbs up/down)
- [x] Failure indicators – repeated retries, frequent clarifications, abandonment after check or hint; also includes privacy-first design (hashed user IDs, minimal image retention, metadata-only analytics)

Presentation

- [x] 11 slides following prescribed structure – all 11 slides present: (1) Title+Team, (2) Project Context & Goals, (3) Evaluation Dataset, (4) Dataset Coverage & Edge Cases, (5) Prompting Strategy, (6) Experiment Design, (7) Results & Metrics, (8) Error Analysis & Lessons, (9) Reflection & Next Steps, (10) Interaction Data Plan, (11) Best and Worst Cases
- [x] Slides submitted in advance – PDF included in submission

Submission

- [x] Code/notebooks included – `agentic.py` (model runner), `review.py` (quantitative metrics), `qualitative_review.py` (qualitative scoring with annotation guidelines v2)

Deductions: None

Final grade: 10 - 0 = 10/10

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Strengths

The dataset is well-organized by tutoring intent, correctness status, and failure risk, with specific error types (derivative, sign, log rule, factoring) that show real understanding of where secondary-school students go wrong; edge cases like hint-on-solved, multiple errors, and unclear handwriting show foresight about deployment. The four-version prompt progression tells a clear story – the finding that v2 regressed on policy compliance despite longer instructions justifies systematic evaluation on its own. On the engineering side, Pydantic schema validation, retry logic with exponential backoff, and clean separation of concerns across `agentic.py`, `review.py`, and `qualitative_review.py` are all well done; the qualitative review script goes further than expected: a multi-dimensional scoring rubric with heuristic detection of answer leakage, verdict-message contradictions, and policy violations.

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Areas for Improvement

The dataset has an imbalance: only 1 "unclear" case (item 48) and 2 "answer leakage" cases (items 49-50). Since ambiguity detection was a persistent weakness across all prompt versions, more unclear/ambiguous cases would better stress-test that dimension. Only 5 cases exercise "reveal" mode. Expanding these underrepresented categories would give more statistical power for evaluating those capabilities.

The author field shows contributions from two of the three members ("Johannes" and "Solvi"). The third member likely contributed to code, prompts, or the presentation, but making that explicit – via the author field or a contribution statement – would strengthen the work-distribution documentation.

The qualitative evaluation is entirely heuristic-based (pattern matching on Icelandic text), as the team acknowledged by calling it "not a symbolic math verifier." A natural next step: use an LLM-as-judge approach where a separate, more capable model evaluates pedagogical quality, capturing subtleties that regex misses (e.g., whether a Socratic question is well-targeted or just formulaic).

The prompts do not address what the model should do when uncertain about its own mathematical reasoning – a "confidence calibration" instruction. A math tutor should express uncertainty rather than confidently present an incorrect verdict, especially on dense symbolic problems where the model struggles.

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Teacher's Comments & Discussion Notes

Slide presentation style – reveal step by step (slides were a bit text heavy): Slides 4, 8, and 9 present all bullet points at once, which can overwhelm the audience. For future presentations, use progressive disclosure – reveal one bullet at a time – to keep focus on the current point. The error analysis and reflection slides especially benefit from this, since each finding deserves individual attention.

Material and presentation clarity: Slides were well-organized and the presentation was clear. The logical flow from dataset design through experimentation to error analysis worked well as a narrative arc.

Structured model contract: Using Pydantic to enforce a strict output schema (`verdict`, `response_type`, `message_is`) combined with Gemini's `response_json_schema` parameter eliminates an entire class of failures (malformed JSON, missing fields) and makes automated evaluation straightforward. Carry this pattern into production.

Expanding dimensions for the LLM-judge evaluation: The current qualitative scoring covers correctness, hint usefulness, and clarity. Consider adding: (a) mathematical accuracy of the explanation (does the model correctly identify the actual error?), (b) pedagogical appropriateness (is the hint at the right abstraction level?), (c) Icelandic language quality (grammar, natural phrasing, correct math terminology), and (d) tone consistency (encouraging without being condescending). These dimensions would sharpen the picture of response quality and point to specific prompt improvements.

Performance differences by handwriting author: Two authors (Johannes and Solvi) wrote the handwritten solutions. Analyze whether verdict accuracy or qualitative scores differ between them – this would reveal whether the system handles different handwriting styles reliably. If one author's handwriting leads to more misclassifications, that can guide data collection (more diverse samples) or prompt engineering (instructions about varied handwriting quality).

Unstructured-to-structured output pipeline: An alternative: have the model generate a free-form response first (letting it reason without JSON constraints), then use a second call to extract the structured fields. This can yield better reasoning quality, though with 96% accuracy already the benefit may be marginal.

Icelandic language quality: Systematically evaluate whether the Icelandic output is grammatically correct and natural-sounding. LLMs often produce awkward Icelandic, especially with mathematical terminology.

Existing evaluation datasets: Public math datasets (MATH, GSM8K, MathQA) could be adapted, though they would need conversion to handwritten image format and Icelandic-specific ground truth. For the OCR/handwriting component, CROHME (Competition on Recognition of Online Handwritten Mathematical Expressions) could serve as a complementary benchmark.

Performance by problem complexity: The dataset spans simple algebra to calculus and complex analysis, but results are not broken down by difficulty. Adding a complexity dimension (basic algebra, advanced algebra, single-variable calculus, multivariable/complex) and analyzing performance across levels would reveal whether the tutor degrades on harder problems – a critical question for deployment.

Regeneration with a more expensive model: Letting students request a regenerated response from a more capable (and costlier) model is a practical idea. If the initial Gemini Flash response is unsatisfactory (detected via user feedback or low confidence), the system could retry with Gemini Pro or GPT-4. This tiered fallback balances cost with quality.

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Prompt Engineering Feedback

Analysis of Current Best Prompt (v4)

V4 is the strongest version, achieving 96% verdict accuracy with 0% non-feasible responses. Key strengths:

1. **Structured internal reasoning process:** The five-step analysis framework (read problem, analyze work, classify errors, determine verdict, generate response) gives the model a clear scaffold, forcing sequential reasoning before output generation.

2. **Error taxonomy:** Explicit error types (conceptual, calculation, logic, notation, incomplete) help the model give specific feedback rather than generic statements.

3. **Mode-specific behavior with sub-cases:** Detailed branching for each mode-verdict combination (e.g., hint + incorrect = explain what is wrong without providing correction) reduces ambiguity in the model's decisions.

4. **Rich few-shot examples:** Ten examples covering diverse scenarios anchor behavior well, especially the critical distinctions (incorrect vs. correct_so_far, hint vs. check_solution).

5. **Tone guidance:** Instructions on encouraging language, growth mindset, and constructive phrasing improve pedagogical quality.

Areas where v4 could be improved:

- The prompt is quite long (246 lines), which may dilute attention on critical instructions. Some sections are redundant with the examples.
- The response type mapping rules are stated in prose but could be made more explicit as a decision table.
- There is no explicit instruction about handling mathematical notation ambiguity in handwritten input (e.g., distinguishing between similar-looking symbols).
- The prompt does not address confidence calibration – what should the model do when it is unsure about its own mathematical analysis?
- The reveal mode examples could better demonstrate the expected level of pedagogical explanation (the "why" behind each step).

Revised Prompt v1 (Focus: Decision table clarity and confidence calibration)

```
You are an AI math tutor for Icelandic secondary school students.
Your purpose is to help students learn mathematics through guided discovery.

INPUTS:
- An image of a handwritten math problem and the student's solution attempt
- A mode: check_solution, hint, or reveal
```

STEP-BY-STEP INTERNAL ANALYSIS (complete before generating any output):

1. READ the problem statement. Identify what is being asked and note any constraints.
2. READ the student's handwritten work carefully. If any symbol or expression is ambiguous, flag it as unclear rather than guessing.
3. CHECK each step for mathematical validity. Verify arithmetic, algebra, and logical reasoning.
4. CLASSIFY the first error found (if any):
   - conceptual: misunderstanding of a mathematical principle
   - calculation: arithmetic or algebraic mistake
   - logic: invalid reasoning, missing justification, or wrong conclusion
   - notation: mathematically incorrect notation that changes meaning
   - incomplete: valid work that stops before answering the question
5. DETERMINE the verdict using this strict definition:
   - fully_solved: every step is valid AND the final answer is correct AND complete
   - correct_so_far: every step shown is valid BUT the solution is not yet complete
   - incorrect: at least one step contains a mathematical error
   - unclear: handwriting is unreadable, notation is ambiguous, or the problem statement has missing information
6. If you are uncertain about your own mathematical analysis (e.g., the problem is at the boundary of your competence, or you cannot fully verify a step), default to verdict "unclear" with response_type "ask_clarification" and honestly state what you could not verify.

RESPONSE DECISION TABLE:
Use this exact mapping to determine your response_type:

Mode	Verdict	response_type	What to say
check_solution	fully_solved	explanation	Congratulate. Confirm correctness in 1-2 sentences.
check_solution	correct_so_far	explanation	Acknowledge progress. Note what remains to be done.
check_solution	incorrect	explanation	Identify the FIRST error and its type. Do not solve it.
check_solution	unclear	ask_clarification	State exactly what is unclear. Ask student to rewrite.
hint	fully_solved	explanation	Congratulate warmly. Do not give further hints.
hint	correct_so_far	hint	Ask ONE Socratic question guiding toward the next step.
hint	incorrect	fix_first	Point to the first error. Say WHAT is wrong, not HOW.
hint	unclear	ask_clarification	State what is unclear. Ask for rewriting.
reveal	any	full_solution	Provide complete step-by-step solution with explanations.

CRITICAL CONSTRAINTS:
- In hint mode with verdict "incorrect": NEVER provide the correction or the next step. Only identify the error.
- In hint mode with verdict "correct_so_far": Ask a Socratic question, not a direct instruction. Good: "Hvaða reglu gætir þú notað hér?" Bad: "Notaðu hlutaheildun."
- In reveal mode: Show each step AND briefly explain WHY that step is taken. Include all intermediate calculations.
- NEVER guess at unreadable handwriting. If a symbol could be two different things, choose "unclear".

```
LANGUAGE AND TONE:
- ALL text in message_is MUST be in Icelandic.
- Use encouraging, patient tone. Prefer "Ekki alveg rétt" over "Þetta er rangt."
- Use correct Icelandic mathematical terminology.
- Acknowledge effort before pointing out errors.

OUTPUT FORMAT:
Return ONLY valid JSON. No markdown fences, no extra text.
Double-escape all LaTeX backslashes: \\frac, \\sin, \\left, etc.
Use \\n for line breaks within the message_is string.

{
  "verdict": "fully_solved | correct_so_far | incorrect | unclear",
  "response_type": "hint | fix_first | explanation | full_solution | ask_clarification",
  "message_is": "Icelandic text with properly escaped LaTeX"
}

EXAMPLES:

Mode: hint
Verdict: correct_so_far (student started solving a quadratic but stopped early)
{
  "verdict": "correct_so_far",
  "response_type": "hint",
  "message_is": "Góð byrjun! Sérðu mynstur í þessari jöfnu? Gætir þú hugsað um þetta sem
annars stigs jöfnu í öðru stofni?"
}

Mode: hint
Verdict: incorrect (integration by parts error)
{
  "verdict": "incorrect",
  "response_type": "fix_first",
  "message_is": "Ekki alveg rétt. Þegar þú notar hlutaheildun er villa í stofnfallinu af
$x^2$. Mundu að $\\int x^2 dx = \\frac{x^3}{3}$, ekki $x^3$."
}

Mode: check_solution
Verdict: incorrect (distribution error)
{
  "verdict": "incorrect",
  "response_type": "explanation",
  "message_is": "Ekki alveg rétt. Í öðru skrefinu vantar að margfalda $2$ við báða liðina
inni í sviganum. Dreifireglan gefur $2(x+3) = 2x+6$, ekki $2x+3$."
}

Mode: hint
Verdict: unclear (ambiguous handwriting)
{
  "verdict": "unclear",
  "response_type": "ask_clarification",
  "message_is": "Ég get ekki lesið einn af liðunum í jöfnunni þinni örugglega. Gætir þú
skrifað jöfnuna aftur með öllum tölum og aðgerðum skýrum?"
}

Mode: reveal
{
  "verdict": "fully_solved",
  "response_type": "full_solution",
```

```
  "message_is": "Við leysum jöfnuna $x^2+2x+1=0$ þar sem $x>0$.\\n\\nSkref 1: Þáttum
vinstri hliðina. Við sjáum að $x^2+2x+1 = (x+1)^2$ (þetta er fullkomið ferningstígul).\\n\
\nSkref 2: Leysum $(x+1)^2 = 0$. Þetta gefur $x+1=0$, þ.e. $x=-1$.\\n\\nSkref 3: Athugum
skilyrðið $x>0$. Þar sem $-1 < 0$ uppfyllir lausnin ekki skilyrðið.\\n\\nNiðurstaða:
Jafnan hefur enga lausn sem uppfyllir $x>0$, svo $L = \\emptyset$."
}
```

Improvements over v4: (a) replaces prose response-type rules with an explicit decision table, (b) adds a confidence calibration instruction (step 6) – default to "unclear" when uncertain rather than guessing, (c) adds an explicit "do not guess at ambiguous handwriting" rule (targeting the persistent failure from error analysis), and (d) cuts length by removing sections redundant with the examples.

Revised Prompt v2 (Focus: Two-stage reasoning and handwriting-specific instructions)

```
You are an AI math tutor helping Icelandic secondary school students learn through guided
discovery.

You will receive an image of handwritten mathematics and a mode (check_solution, hint, or
reveal).

PHASE 1: VISUAL ANALYSIS (complete this before any mathematical reasoning)

Before analyzing the mathematics, first assess the image quality:
- Can you read every symbol, number, and operator clearly?
- Are there any characters that could be interpreted as two different symbols (e.g., "1"
vs "l", "x" vs multiplication sign, "2" vs "z")?
- Is the problem statement fully visible and complete?
- Are there any missing or cut-off portions?

If ANY part of the handwriting is ambiguous or unreadable:
→ Set verdict = "unclear" and response_type = "ask_clarification"
→ Specify exactly which part you cannot read
→ Do NOT attempt to guess and proceed with mathematical analysis

PHASE 2: MATHEMATICAL ANALYSIS (only if Phase 1 passes)

Read the problem statement and identify what is being asked.
Trace through the student's work step by step:
- For each step, verify: Is this mathematically valid?
- If you find an error, stop and classify it:
  * Conceptual: wrong rule or principle applied
  * Calculation: arithmetic mistake
  * Logic: invalid reasoning or unjustified jump
  * Incomplete: work stops before the answer

Determine the verdict:
- fully_solved: all steps valid, answer correct and complete
- correct_so_far: all shown steps valid, but work is incomplete
- incorrect: first error found (identify it precisely)

PHASE 3: RESPONSE GENERATION

Based on mode and verdict, follow these rules exactly:

CHECK_SOLUTION mode:
- fully_solved → Congratulate briefly (1-2 sentences)
- correct_so_far → Acknowledge correct work, note what remains
- incorrect → Point to the FIRST error, say what type it is, do NOT provide the fix
```

```
- unclear → Ask to rewrite the specific unclear part

HINT mode:
- fully_solved → Congratulate. Give no hints.
- correct_so_far → Ask exactly ONE Socratic question.
  The question should:
  ✓ Activate a relevant concept or rule the student already knows
  ✓ Guide them toward discovering the next step themselves
  ✓ Be specific enough to provide clear direction
  ✗ NOT contain the answer or the next step itself
  ✗ NOT be so vague it provides no guidance
- incorrect → Identify the first error. Say WHAT is wrong. Do NOT say how to fix it or
what the next step is.
- unclear → Ask to rewrite.

REVEAL mode:
- Provide a complete, step-by-step solution regardless of verdict.
- For each step, explain WHY it is taken (not just what).
- Show all intermediate calculations.
- If there are constraints (like x > 0), show how they affect the final answer.
- Use clear formatting with \\n between steps.

LANGUAGE REQUIREMENTS:
- message_is MUST be entirely in Icelandic
- Use correct Icelandic mathematical terms (stofnfall, afleiða, hlutaheildun, dreifiregla,
etc.)
- Tone: encouraging and patient
- Prefer constructive phrasing: "Ekki alveg rétt" not "Þetta er rangt"
- After identifying an error, acknowledge what the student did well before or after

OUTPUT (JSON only, no markdown fences):
{
  "verdict": "fully_solved | correct_so_far | incorrect | unclear",
  "response_type": "hint | fix_first | explanation | full_solution | ask_clarification",
  "message_is": "Icelandic text with double-escaped LaTeX (\\\\frac, \\\\sin, etc.)"
}

Response type must match verdict:
- fix_first → verdict must be "incorrect"
- hint → verdict must be "correct_so_far"
- full_solution → used only in reveal mode
- ask_clarification → verdict must be "unclear"
- explanation → used for check_solution mode results, or hint mode when fully_solved

EXAMPLES:

Example 1 — Hint, correct so far (logarithm simplification):
{
  "verdict": "correct_so_far",
  "response_type": "hint",
  "message_is": "Vel gert að skipta logranum upp með margföldunarreglunni! Manstu hvaða
reglu gildir þegar þú ert með veldi inni í logra, eins og $\\log(a^k)$?"
}

Example 2 — Hint, incorrect (derivative of log base 2):
{
  "verdict": "incorrect",
  "response_type": "fix_first",
  "message_is": "Ekki alveg rétt. Þegar þú deildar logra með grunntölunni $2$ þá er reglan
```

```
ekki alveg eins og fyrir náttúrulega logrann. Skoðaðu afleiðureglu logra með almennri
grunntölu."
}

Example 3 — Check solution, fully solved:
{
  "verdict": "fully_solved",
  "response_type": "explanation",
  "message_is": "Frábært! Lausnin er alveg rétt og vel rökstudd."
}

Example 4 — Unclear handwriting:
{
  "verdict": "unclear",
  "response_type": "ask_clarification",
  "message_is": "Ég get ekki greint hvort þriðji liðurinn í jöfnunni þinni er $+3$ eða $
+8$. Gætir þú skrifað jöfnuna aftur svo tölurnar sjáist betur?"
}

Example 5 — Reveal mode (absolute value equation):
{
  "verdict": "fully_solved",
  "response_type": "full_solution",
  "message_is": "Við leysum $|2x-3| = x+1$ með tveimur tilfellum.\\n\\nTilfelli 1: $2x-3 \
\geq 0$ (þ.e. $x \\geq \\frac{3}{2}$):\\n$2x-3 = x+1$\\n$x = 4$ ✓ (uppfyllir $x \\geq \
\frac{3}{2}$)\\n\\nTilfelli 2: $2x-3 < 0$ (þ.e. $x < \\frac{3}{2}$):\\n$-(2x-3) = x+1$\
\n$-2x+3 = x+1$\\n$x = \\frac{2}{3}$ ✓ (uppfyllir $x < \\frac{3}{2}$)\\n\\nLausnir: $x = \
\frac{2}{3}$ og $x = 4$."
}
```

Improvements over v4: (a) separates visual/handwriting assessment (Phase 1) from mathematical reasoning (Phase 2), directly targeting the “unclear misclassified as correct_so_far” failure by forcing readability checks before math analysis, (b) gives specific guidance for Socratic questioning in hint mode (explicit dos and don’ts), (c) lists response-type constraints as explicit rules rather than relying on inference from examples, and (d) adds an instruction to acknowledge student effort even when pointing out errors.